Assignment - 4

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Section - A.

Que 1- What is transaction in DBMS?
Ans 1- A transaction is a logical unit of work that performs multiple database operations like insert, update, or delete. These operations are treated as a single unit to maintain data integrity. A transaction must either complete fully or have no effect at all.

Que 2- Explain ACID properties.
Ans 2- ACID stands for Atomicity, consistency, isolation, and durability. These properties ensure that transactions are processed reliably and the database remains accurate. They prevent data loss and inconsistency during failures or concurrent execution.

Que 3- What is concurrency ~~mo~~ control?
Ans 3- Concurrency control is a method of managing multiple transactions executing at the same time. It ensures that simultaneous operations do not create conflicts or corrupt data. It maintains consistency in multi-user environments.

Que 4- What is deadlock?
Ans 4- A deadlock occur when two or more transactions

are waiting for each other to release resources. As a result, none of them can proceed further. The situation can freeze the system if not handled properly.

Que 5- What is serializability?
Ans 5- Serializability ensures that the results of concurrent transactions is the same as if they were executed one after another. It is a key concept in maintaining correctness during concurrency. It helps avoid anomalies in the database.

Que 6- What is recovery in DBMS?
Ans 6- Recovery is the process of restoring the database to a consistent state after a system crash or failure. It uses logs or backup information to undo or redo transactions. This ensures no data is permanently lost.

Que 7- What is checkpoint?
Ans 7- A checkpoint is a saved state of a database at a particular time. It reduces the amount of work needed during recovery after a crash. The system can restart from the last checkpoint instead of beginning.

Section-B

Que 1- Explain ACID properties using banking example.
Ans 1- When money is transfered from one account to another, both debit and credit operations must occur together. The total balance remains consistent, no other user see partial updates, and the changes are saved

permanently after completion.

Que 2- Illustrate a deadlock situation with two transactions.

Ans 2- If transaction T1 locks Account A and waits for Account B, while T2 locks B and waits for A, both transactions keep waiting. This creates a deadlock where neither can proceed.

Que 3- Explain serial vs concurrent execution with example.

Ans 3- In serial execution, transactions run one after another, which is safe but slow. In concurrent execution, multiple transactions run at the same time, which improves speed but requires control to avoid conflicts.

Section C

Que 1- A banking system transfers money between accounts. Explain how ACID properties apply.

Ans 1- During money transfer, atomicity ensures both steps occur, consistency keeps balance correct, isolation prevents interference, and durability saves changes permanently. This maintains trust in the system.

Que 2- Two transactions update same data simultaneously. Identify possible issues.

Ans 2- This can cause problem like lost updates, dirty reads or inconsistent data. Proper locking and concurrency control techniques are required to handle this situation.

Que3- A system crashes during transaction. Explain recovery steps
Ans3- The system checks the log file after a crash. If a transaction was committed, it is redone; if not committed, it is rolled back. this restores the database to a consistent state.

Section-D.

Que1- Explain one ACID property with real-life.
Ans1- Atomicity means a transaction must be completed fully or not executed at all. For example, in online ticket booking, the seat reservation and payment both happen together! If payment fails, seat is not booked This avoids partial updates and keeps the system reliable.

Que2- Draw a transaction and identify conflict.
Ans 2- T1: Read(A) → Write(A)
T2: Read(A) → Write(A)

Here, both transactions are accessing and modifying the same data item A at the same time. This create a write-write conflict and may lead to incorrect or overwritten values. Proper concurrency control like locking is needed to avoid this issue.

Que3- Explain one recovery technique.
Ans3- Log-based recovery uses undo and redo operation to restore the database. It ensures committed transactions are saved and incomplete ones are removed.